

EFFECT OF PROGESTERONE ON GROWING CARTILAGE AND BONE IN IMMATURE GUINEA PIGS

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In continuation of investigations of the effect of various hormones on the changes of cartilage and bone taking place during growth and aging, the present study is concerned with the response of the skeletal tissues of growing guinea pigs to progesterone.

MATERIAL AND METHODS

Twenty guinea pigs, 10 females and 10 males, weighing 150 to 170 Gm. at the beginning of the experiment, received subcutaneous injections of progesterone in the form of proluton (Schering) in sesame oil on six days of the week for periods of one, two, four, six and eight weeks. In one half of the animals the dose given each time was 1 rabbit unit, and in the other half, 0.65 rabbit unit.

Frontal sections were prepared from the upper third of one tibia, whereas the knee joint of the other side, which had been left intact at autopsy, was sectioned in a sagittal direction. Longitudinal sections were also made of the chondro-osseous junction of several ribs.

GENERAL OBSERVATIONS

Sixteen animals gained weight in a normal manner, the weights rising from 150 to 395 Gm. during a period of eight weeks, which is according to expectation. At autopsy the long bones, as well as the ribs, appeared thinner and more delicate than ordinarily. The length of the tibia, taken with the aid of a caliper, was 4.05 to 4.2 cm. after eight weeks of treatment, which is about normal.

Three guinea pigs of a group of 4 showed, after six weeks of injections, a sudden drop in weight from about 340 to 250 Gm., without apparent cause. These animals were therefore killed six and a half weeks after the beginning of the experiments. No specific lesions indicating intercurrent diseases were found at autopsy. A female belonging to another group likewise failed to gain weight during two weeks' observation.

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MICROSCOPIC OBSERVATIONS

After progesterone had been administered for one or two weeks the *zone of endochondral ossification* of the upper part of the tibia was somewhat enlarged. The chondromucoid ground substance was scarce, and its fibrils were delicate. The cartilage cell rows had a regular structure. A single row of these cells contained, as a rule, 12 columnar and 4 hypertrophic cartilage cells. The columnar cells were thus slightly increased over the normal number of 10, whereas the number of hypertrophic cells was unchanged. The nonoriented, so-called resting cartilage cells were enlarged and sent out numerous branching processes. The ordinarily flat columnar cartilage cells took on a polyhedral or ellipsoid shape. This was particularly noticeable in the proximal parts of the cartilage columns. Both the nonoriented and, in particular, the enlarged columnar cartilage cells proliferated actively and frequently by mitosis. The hypertrophic cartilage cells were large, and less calcium salts were deposited at their periphery than is normal. The capsules of the hypertrophic cartilage cells were readily opened by capillaries advancing from the bone marrow. The demarcation between the layer of hypertrophic cartilage cells as a whole and the bone marrow was sharp.

If the injections were continued for a period of one month, the amount of cartilaginous ground substance in the growth zone was still small, although the matrix was slightly swollen; normally at this age the cartilaginous ground substance has begun to increase in amount and becomes dense and sclerosed. The cellular structure of the epiphyseal disk did not differ very much from that seen after one or two weeks of treatment with progesterone. Therefore in appearance the epiphyseal disk of guinea pigs which had been treated with progesterone for one month and weighed 250 to 285 Gm. was comparable to that of untreated animals which were about one month younger and which had a weight of 150 to 180 Gm.

After administration of progesterone for a period of two months, the guinea pigs weighed between 365 and 396 Gm. The zone of endochondral ossification in these animals was still of good medium width, whereas normally guinea pigs of similar age and weight already show narrowing of the epiphyseal disks. The amount of cartilaginous ground substance in the epiphyseal growth zone was only slightly increased as compared with that at earlier stages and much smaller than normal. It also failed to show the increasing hyalinization and sclerosis usually seen at this age. The cell count in a single cartilage cell row was only a little higher than normal, namely, 3 to 4 hypertrophic and 9 to 10 columnar cartilage cells. But the cells were conspicuously larger, and more cell rows were present than in corresponding control animals (fig. 1 *A* and *B*). The nonoriented and the columnar cartilage cells were still observed to proliferate, and here and there they underwent mitotic division, a condition not found at this age under ordinary circumstances. The hypertrophic cartilage cells, although more calcified than at the earlier stages, were larger and showed somewhat less calcification than those in normal control guinea pigs, and bone formation was likewise decreased. Furthermore, while in normal guinea pigs of corresponding age some cartilage cells showed retrogressive changes and the onset of the closure of the growth zones was indicated by the formation of osseous plugs, in animals treated with progesterone no atrophy or degeneration of single cartilage cells or of whole cell rows and no formation of osseous plugs was observed. Thus, after two months of treatment the microscopic appearance of the epiphyseal cartilage of animals weighing then about 400 Gm. was comparable to that of untreated guinea pigs four to six weeks younger and weighing about 250 Gm.

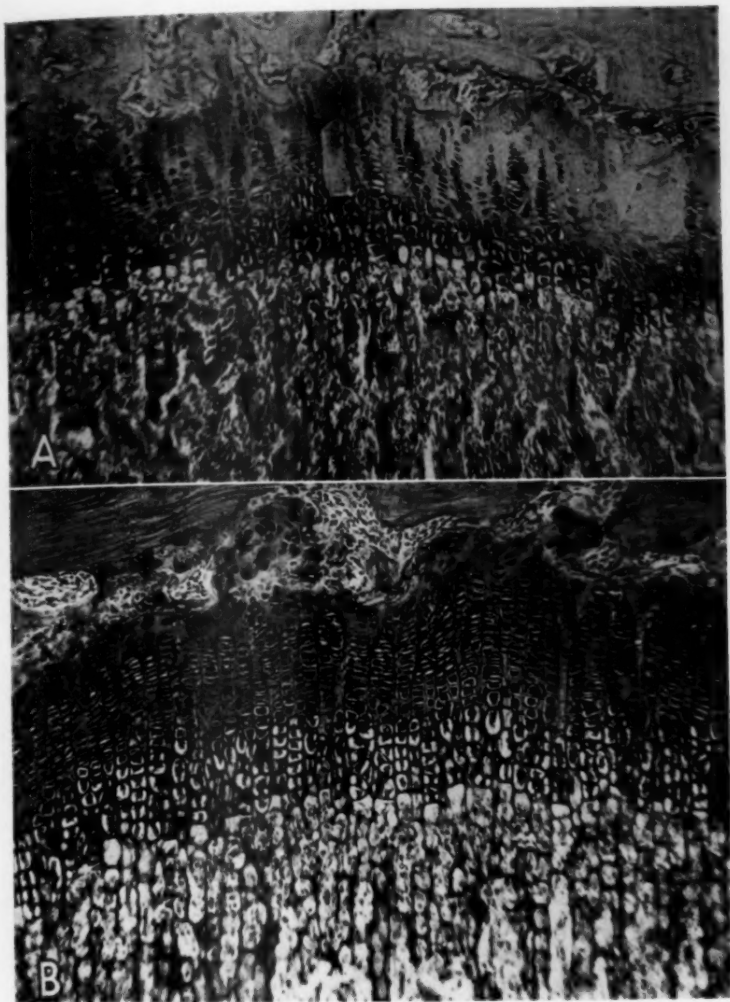


Fig. 1.—*A*, section through the epiphyseal disk of the upper tibia of a normal female guinea pig whose weight during eight weeks of observation had risen from 150 to 405 Gm. Fairly large amounts of cartilaginous matrix are present; columnar cartilage cells are small and somewhat irregularly arranged. In the center of the disk an area of localized ossification is seen. Magnification, $\times 150$.

B, section through the epiphyseal disk of the upper tibia of a female guinea pig which had received 1 rabbit unit of progesterone six times weekly for a period of eight weeks. The weight of the animal had increased from 150 to 395 Gm. during this period. Little ground substance is present, and the cartilaginous disk is wider than in *A*. The columnar and hypertrophic cartilage cells are more numerous and larger than normally. The formation of subepiphyseal trabeculae is lagging behind that seen in *A*. Magnification, same as in *A*.

The subepiphyseal zone was wide and well vascularized. In the early stages relatively few osteoblastic cells were present, and their conversion into osteocytes took place at a slow rate. After one month of injections of progesterone, large osteoblastic epithelioid cells were more numerous. These proliferated mitotically and were lying close together in and around corroded cartilage cell capsules. Farther distally they were situated along the longitudinal processes, which were remnants of the calcified walls of corroded cartilage cell capsules. Thus trabeculae were formed. But, because little osseous ground substance was laid down, the conversion of the osteoblasts into dense and mature small osteocytes was delayed. A soft fibrocytic connective tissue interspersed with some bone marrow cells separated these thin and rather immature subchondral spicules from each other. Few osteoclastic giant cells were seen. Farther distally in the marrow cavity the bone marrow was cellular and contained fairly numerous but thin bony trabeculae, at the surface of which osteoblasts, as well as some osteoclastic giant cells, were present.

The findings described so far were observed in 16 of 20 guinea pigs. In 4 animals, which had lost or failed to gain weight, a reaction different from that seen in the majority of the animals was noted microscopically. These 4 had certain histologic features in common. The growth zones were narrowed. The cartilage cells were decreased in number and size. The cell count in a single cartilage cell row had decreased to 1 to 2 hypertrophic and 6 columnar cells. The cartilaginous matrix was increased in amount and dense, and a collagenous and preosseous layer of ground substance traversed the narrowed zone of the hypertrophic cartilage. The proliferation of the cartilage cells continued, and here and there mitotic division of columnar cartilage cells was seen. The subepiphyseal layer was poorly vascularized and narrowed, and bone formation took place close to the zone of hypertrophic cartilage and was intensified as compared with conditions in those guinea pigs which, under the influence of progesterone, had not shown a loss in weight. The histologic appearance of the epiphyseal zones of those guinea pigs which had lost weight resembled, therefore, that seen in underfed guinea pigs.¹ Yet the influence of the progesterone was recognizable to a certain extent, as the epiphyseal cartilage cells were larger and more numerous and less bone was present than in underfed guinea pigs which had not been treated with progesterone. Atrophy and solution of the cells of the bone marrow also indicated that in these animals there might have been certain nutritional disturbances.

Joints.—After one or two weeks of injections of progesterone the ordinarily flat cells of the upper layer, composing the sliding zone, enlarged, rounded off and became more numerous. The cells of the transitional and pressure zones also increased in size and not infrequently showed mitotic proliferation. The cartilaginous ground substance was diminished in amount. The hypertrophic cartilage cells were less calcified than ordinarily; they readily broke down under the influence of elements of the bone marrow. The subchondral tissue contained large osteoblasts, the conversion of which into osteocytes was delayed. These changes became more accentuated four, and particularly eight, weeks subsequent to the beginning of the administration of progesterone. In the sliding and transitional zones numerous cells were present, which still proliferated here and there by way of mitosis. These cells were larger and the formation of cartilage capsules was much less distinct than usual (fig. 2A and B). Whereas in normal guinea pigs of corresponding age and weight the trabeculae consisted of dense and lamellated

1. Silberberg, M., and Silberberg, R.: Arch. Path. 30:675, 1940.

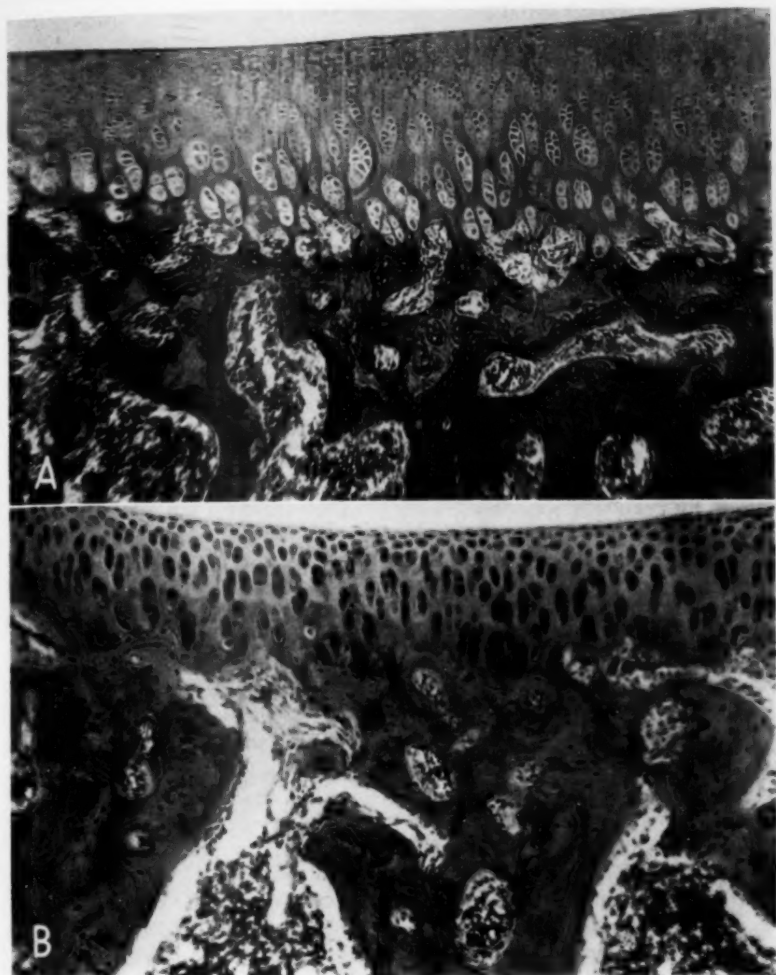


Fig. 2.—*A*, section through the upper articular surface of the tibia of a normal female guinea pig whose weight had risen from 150 to 380 Gm. during eight weeks of observation. The various cartilage cell layers can be readily distinguished. The bony trabeculae consist of dense ground substance and small osteocytes. Magnification, $\times 220$.

B, section through the upper articular surface of the tibia of a female guinea pig which had received 0.65 rabbit unit of progesterone six times weekly for a period of eight weeks. The weight of the animal had increased from 160 to 385 Gm. during this period. Cells of the various layers are somewhat hyperplastic and hypertrophic. Cartilage capsules are less distinct than in *A* and there is less ground substance present. The bony trabeculae contain large osteocytes and soft ground substance. Magnification, same as in *A*.

bony substance in which small osteocytes were situated, under the influence of progesterone the subchondral trabeculae contained large osteocytes separated from each other by a good deal of soft osseous substance. Only small amounts of calcium had been laid down in the bony substance, and the lamellation of the latter was less sharp. The bony tissue therefore appeared rather immature.

In the 4 animals which had lost or failed to gain weight, the matrix of the articular cartilage was increased in quantity and sclerosed. The cartilage cells were smaller and less numerous and more bone had been produced in the subchondral layer than in those guinea pigs which had shown a steady gain in weight. Thus the conditions were comparable to those in the epiphyseal zones of these particular guinea pigs, and the influence of the nutritional disturbance manifested itself also in the articular cartilage. But, as was the case in the growth zones, a certain effect of the progesterone was noticeable even here, as the number and size of cells were larger than in underfed animals which had not been treated with progesterone.

Bony Shaft.—The changes which took place in the compact bone of the shaft under the influence of progesterone were similar to those seen in the osseous trabeculae. Increasingly during the first four weeks, and still more so after two months of treatment, the compacta became less firm than ordinarily and showed wide haversian canals. The osteocytes and their nuclei were large, and between the bone cells were found fairly large amounts of a soft and only slightly calcified ground substance. The periosteal osteoblastic epithelioid cells which were in contact with the outer surface of the shaft were less numerous than normally; osteoclastic giant cells were even scarcer. The periosteal connective tissue was soft and cellular. No changes were noted in the cells lining the inner surface of the compacta.

In the guinea pigs which showed a loss or lack of gain in weight, the compact bone was denser and more intensely calcified. Likewise the periosteal connective tissue was dense and fibrillar, and the osteoblastic epithelioid cells were more numerous but smaller than those in the animals which had gained in weight.

Ribs.—After administration of progesterone the growth zone at the chondro-osseous junction of the ribs was enlarged. The columnar cartilage cells were more numerous and larger than ordinarily. The hypertrophic cartilage cells contained less calcium than normally. As was the case in the long bones, the osseous structures were less mature than those in control animals. Even as late as eight weeks after the beginning of the injections the cartilaginous ground substance was scarce and showed little sclerosis and hyalinization. The cartilage cells were better preserved than under normal conditions and manifested fewer retrogressive changes than is usual for this age.

In those guinea pigs which had not gained or which had even lost weight, the growth zone was rather narrow. The processes of calcification, hyalinization and sclerosis of the cartilage were increased, and more bone had been produced than in the animals which had gained weight steadily.

COMMENT

In growing guinea pigs, progesterone administered for periods up to two months exerts the following effects on cartilage and bone: (1) It slightly promotes the proliferation and increases the hypertrophy of the epiphyseal and articular cartilage cells; (2) it inhibits retrogressive changes, calcification and ossification of the epiphyseal cartilage, which

would normally occur during the period of observation, and it thus maintains the juvenile character of this tissue for an abnormally long period; (3) it inhibits to some extent the deposition of dense osseous substance and the conversion of immature osseous structures into mature bone; (4) in the ribs it inhibits atrophy, degeneration and calcification and thus counteracts the age changes of the costal cartilage.

These effects bear a certain similarity to those taking place in cartilage and bone subsequent to ovariectomy.² They differ from the latter particularly in that the active proliferation of the cartilage and the resorptive processes involving the bony structures are more pronounced in ovariectomized guinea pigs than in animals treated with progesterone.

As to the effect of progesterone on the cartilage, the slight hyperplasia and the increase in hypertrophy of the cartilage cells may be interpreted as actual stimulation of these cells by the administered substance. Besides this, however, there seems to be another and more pronounced effect, which consists in an inhibition of the normal processes of ossification and of aging.

While the stimulation of growth may be directly due to progesterone, there is also the possibility that this substance may act on the skeletal tissues through the intermediation of other endocrine glands. With reference to the inhibition of the age changes, it is conceivable that the antagonism which Loeb,³ Courrier,⁴ Allen and Meyer⁵ and others have shown to exist between estrogen and progesterone in the uterus and vagina may in some way be involved in the action of progesterone on cartilage and bone. The changes in the skeleton observed subsequent to the administration of progesterone represent almost the opposite of the acceleration of the age changes which we found in the skeletal tissues following injections of estrogen.⁶ An inhibition of the effect of estrogen, produced in the animals treated with progesterone, may partly account for the similarity of the findings subsequent to ovariectomy and to administration of progesterone. But while progesterone inhibits the effects of estrogen, ovariectomy eliminates them, provided other organs do not take over the production of the latter substance.

The duration of our experiments was comparatively short. We cannot, therefore, decide at present whether the effects exerted by progesterone on the skeleton are permanent or transitory or whether they

2. Silberberg, M., and Silberberg, R.: *Am. J. Path.* **16**:491, 1940.

3. Loeb, L.: *J. Morphol.* **22**:37, 1911; *Am. J. Anat.* **32**:305, 1923. Loeb, L., and Kountz, W. B.: *Proc. Soc. Exper. Biol. & Med.* **24**:728, 1927; *Am. J. Physiol.* **84**:283, 1928. Moore, E.; Smadel, J. E.; Koenig, K. F., and Loeb, L.: *Growth* **1**:119, 1937.

4. Courrier, R.: *Compt. rend. Soc. de biol.* **104**:280 and 1178, 1930.

5. Allen, W. M., and Meyer, R. K.: *Anat. Rec.* **61**:427, 1935.

6. Silberberg, M., and Silberberg, R.: *Arch. Path.* **28**:340, 1939.

may even be reversed at later stages, a condition which has been observed to occur subsequent to administration of bovine hypophysial extract and of thyroxin.⁷

SUMMARY

In growing guinea pigs, progesterone administered subcutaneously over periods of up to two months (1) increases slightly the proliferation and more markedly the hypertrophy of the euhyaline cartilage of the epiphysial zones, articular surfaces and ribs; (2) inhibits the retrogressive changes, calcification and sclerosis of the cartilage which normally occur with advancing age, and (3) inhibits the conversion of immature osseous substance into mature bone, thus counteracting the aging of bone.

No definite conclusions can as yet be drawn as to the mechanism underlying these changes—in particular, as to whether all or some of these changes are due to a direct effect of progesterone on the skeletal tissues or whether an intermediary action of other endocrine glands or hormones is involved.

7. Silberberg, M., and Silberberg, R.: *Am. J. Path.*, to be published; *Growth*, to be published.